

Education

Technological Institute of the Philippines

Quezon City, MM

Bachelor of Science in Computer Engineering

Specializes in Systems Administration focused on Linux Systems

August 2025

Graduated with Academic Distinction – GWA: 1.38

Relevant Coursework: Systems Administration; Managing Enterprise Servers; Automation Scripting

Pasig Catholic College

Pasig City, MM

Graduated with High Honors

2021

Microsoft Excel Specialist

Experience

TWIRELESS Inc.

Pasig City, MM

Hardware and Technical Support Intern

March 2025 – June 2025

- Collected and analyzed system performance data to troubleshoot and optimize hardware/software usage
- Created reports and maintained logs for technical issues, improving trend tracking and decision-making.
- Assisted in asset and inventory management using Excel and manual tracking, reducing discrepancies by streamlining processes.
- Collaborated with multiple departments to deliver technical solutions, improving operational efficiency.

Pasig City Hall

Pasig City, MM

Special Program for the Employment of Students - Administrative Support Assistant

2021

- Assisted with document processing in the Pasig City Real Property Tax Office, streamlining the workflow and reducing processing time.
- Aided residents regarding real property tax inquiries, ensuring accurate and timely information delivery.

Projects

Design of a Web-based Monitoring Application for Adaptive Fish Silage Fermentation Control System (2025)

- Designed and developed a data-driven monitoring application using React.js, Firebase, and Koa.
- Implemented real-time ETL pipeline: collected pH, ammonia, and temperature sensor data via ESP32 and processed through Firebase
- Built visual dashboards with indicators for actionable decision-making.
- Help achieve real-time monitoring accuracy with minimal latency.

Design of an Adaptive Fish Silage Fermentation Control System (2025)

- Developed a React Native mobile app with Firebase and Koa for real-time monitoring and control.
- Calibrated and integrated sensors (load-cell, pH, temperature, ammonia) for accurate data collection.
- Structured and managed the data pipeline, ensuring reliable ETL and synchronization between hardware, database, and application.

Development of A Portable Solar-Powered Desalination System with Integrated Ultra-filtration for Safe Drinking Water (2024)

- Proposed the core idea and led the design of the portable and modular desalination system.
- Designed modular system housing for data-logging sensors.
- Implemented PPM sensor logic with real-time data validation for water safety.

Skills & Interest

Technical: Proficient in Linux system administration, Git, Ansible, and automation scripting (Bash, shell); strong understanding of network architecture and router/switch security; experienced with Embedded Systems/IoT (Arduino, ESP32); basic proficiency in SQL using MySQL; familiar with Scrum methodology and the Software Development Life Cycle (SDLC); applies object-oriented programming (OOP) in C++, Python, and JavaScript.

Programming Skills: Working knowledge of C++, Python, and JavaScript (OOP); basic proficiency in SQL using MySQL; experience using Firebase Realtime Database; basic knowledge of Java.

Certifications: Microsoft Office Specialist (2020), AWS Academy Cloud Foundations (2024), & Cisco Devnet Associate (2025)

Awards: Pasig City Scholar (2022–2025); President's Lister (2021–2023); VPAA Lister (2022–2025); Dean's Lister (2023–2024)